MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11745240
SCORING-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Scoring-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11745240	Circuit Card Assembly, Scoring-3

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item identification.- This assembly is the third (3) of three (3) single ended printed circuit boards consisting of RMS to DC converters, op-amp, and divider necessary to provide scoring averages and recticle information.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745240.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with the following power supplies:

a. +10 VDC
b. +15 VDC
c. -15 VDC
d. (0 to +5) VDC
e. (o to -5) VDC

3.2.1.2 AC to DC Converter #1.- Converts input Pin 26 RMS sinusoidal voltage to output Pin 30 DC voltage.

3.2.1.3 AC to DC Converter #2.- Converts input Pin 24 RMS sinusoidal voltage to output Pin 4 DC voltage.

3.2.1.4 Analog Voltage Divider Network.- The analog voltage divider network shall receive the input Pin 34 signal and yield the output on Pin 32.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11745240.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11745240 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

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4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (See Figure I)

4.2.1.1 Power Supplies.- The power supplies shall be as follows:

a. +15.0 VDC ± 5 mVDC, 100 mA or greater and a ripple of less than 10 mV RMS.
b. +20.0 VDC ± 5 mVDC, 100 mA or greater and a ripple of less than 10 mV RMS.
c. -20.0 VDC ± 5mVDC, 100 mA or greater and a ripple of less than 10 mV RMS.
d. (0.000 to +10.000) VDC ± 3mVDC, 100 mA or greater and a ripple of less than 10m V RMS.
e. (0.000 to -10.000) VDC ± 3mVDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Sinusoidal Generator.- The generator shall be capable of producing a 1 KHz ± 5 Hz sine wave having a 2 V ± .005 V peak to peak amplitude.

4.2.1.3 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanoseconds or less, sweep period of 50 nanoseconds per centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.4 Dummy Loads.- The dummy loads shall be three 10 K ohms resistors.

4.2.1.5 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these test shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All requirements shall be met with the following power supplies:

a. +10 VDC ± 5 mVDC
b. +15 VDC ± 5 mVDC
c. -15 VDC ± 5 mVDC
d. (0.000 to +5.000) VDC ± 3m VDC
e. (0.000 to -5.000) VDC ± 3m VDC

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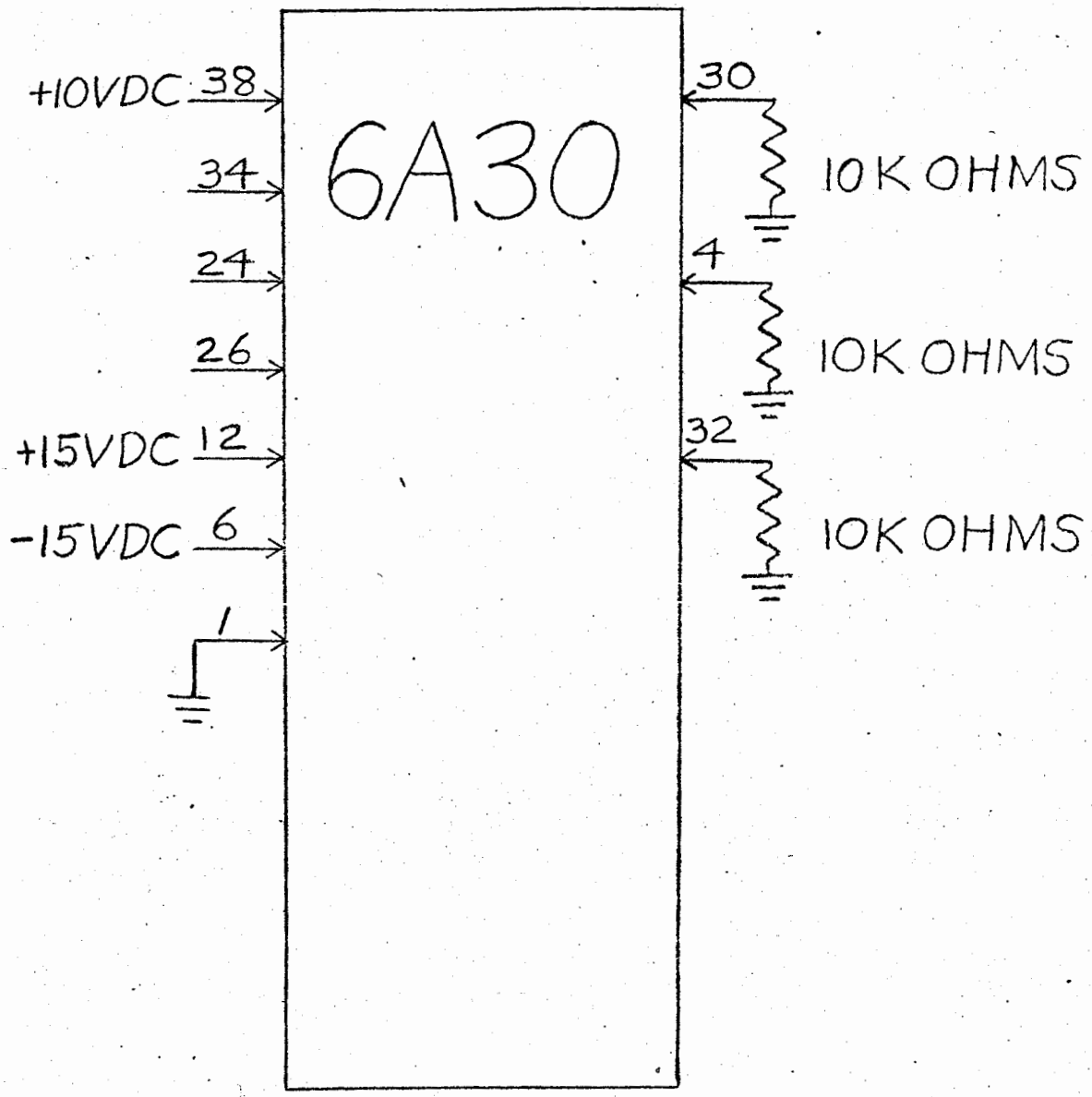


FIGURE I

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4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-Up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 AC to DC Converter #1 Test.- With a 2 V ± .005 V peak-peak, 1 KHz ± 5 Hz sine wave applied to Pin 26, the output on Pin 30 shall be +.707 VDC ± .050 VDC.

4.2.2.3.2 AC to DC Converter #2 Test.- With a 2 V ± .005 peak-peak, 1 KHz ± 5 Hz, sine wave applied to Pin 24, the output on Pin 4 shall be +.707 VDC ± .050 VDC.

4.2.2.3.3 Analog Voltage Divider Network Test.- The analog voltage divider network shall divide the input voltage applied to Pin 34 and yield the output on Pin 32, after having initially adjusted the divider, as shown in Table 1.

NOTE: Initial adjustment.- Apply ground to Pin 34, 10 VDC ± 5 mVDC to Pin 38, and adjust R5 until 5 VDC ± 5 mVDC appears at Pin 32. Remove ground from Pin 34 and vary the applied voltage as shown in Table I, reading output Pin 32 after each input setting.

TABLE I

Input (VDC) Pin 34	Output (VDC) Pin 32
1.0 V ± 3 mV	4.00 ± .025
2.0 V ± 3 mV	3.00 ± .025
3.0 V ± 3 mV	2.00 ± .025
4.0 V ± 3 mV	1.00 ± .025
5.0 V ± 3 mV	0.00 ± .025
-1.0 V ± 3 mV	6.00 ± .075
-2.0 V ± 3 mV	7.00 ± .075
-3.0 V ± 3 mV	8.00 ± .075
-4.0 V ± 3 mV	9.00 ± .075
-5.0 V ± 3 mV	10.00 ± .075

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745240.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

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Preparing Activity:

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Project No. 1240-A620

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